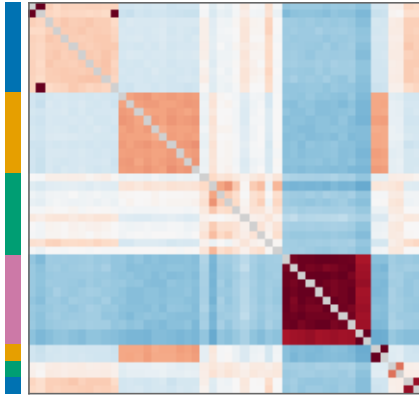
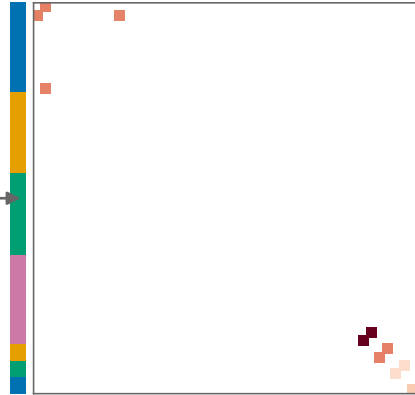


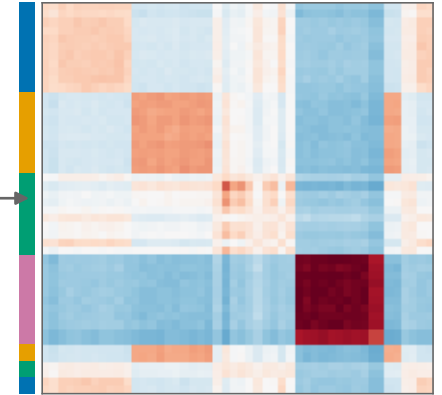
1 one pass: GRM C
diagonal unobserved (grey)



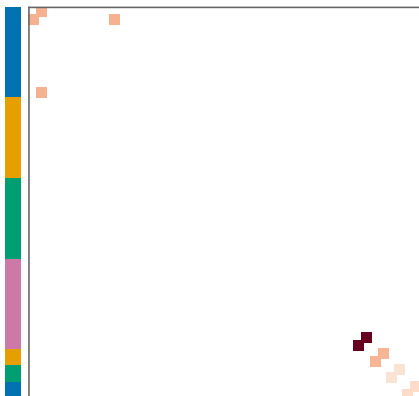
2 KING-robust > 0.04:
the only pairs allowed in S



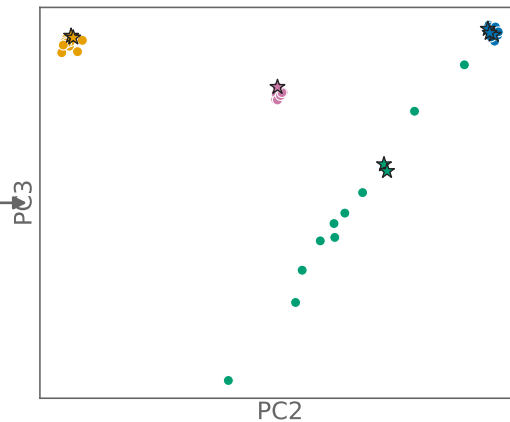
3 PCP $C = L + S$:
structure L, diagonal imputed



4 S: related pairs
($\hat{\phi} > \tau$, candidates only)

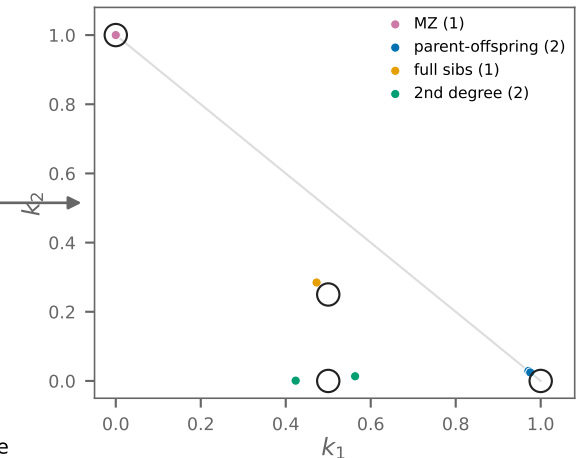


5 PCs of L (--impute-diag)
relatives as stars



● CEU ● CHB ● MXL ● YRI ★ relative

6 final relatedness (1 pass)
IAF from the PCs; o = expected



All $N \times N$ in memory after one pass; each PCP iteration needs a full eigendecomposition ($N = 500$: 7.5 s; $N = 1000$: 33 s).
No HWE assumption and no K . At small N the observed diagonal reflects heterozygosity, not structure: use --impute-diag.
Example: admixTjeck2, $N = 48$ with 8 relatives.